Dev Marwah

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Education

Vellore Institute of Technology, Bhopal — Bhopal, MP
B.Tech in Computer Science — Specialization in AI & ML — *Sept. 2023 – Present*

Jesus And Mary Sr. Sec. School — Bikaner, RJ
Senior Secondary — *2021 – 2023*

Delhi Public School — Bikaner, RJ
Schooling — *2010 – 2021*

Projects

SpringBoot E-Commerce | ***Java, Spring Boot, Spring Data JPA, MySQL, React*** — June 2026 | Live | GitHub

- Built a full-stack e-commerce REST API with Spring Boot and Spring Data JPA, supporting full CRUD operations (add, search, update, delete) and multipart image upload/storage.
- Implemented a keyword-based product search endpoint using JPA derived queries, returning fast, real-time results as the user types.
- Deployed the backend on Render with a managed Aiven MySQL instance and the React frontend on Vercel, using environment-based configuration to fully externalize local vs. production database credentials.
- Debugged and resolved Spring Boot 4.x breaking changes during migration, ensuring compatibility across the persistence layer and REST controllers.
- Structured the API layer with clean separation between controllers, services, and repositories, making the codebase straightforward to extend with new product and order features.

Handwritten Digit Recognition from Scratch | ***Python, NumPy*** — Jan 2024 | GitHub

- Built a fully connected neural network in Python without ML frameworks to classify handwritten MNIST digits, achieving 95%+ accuracy.
- Implemented forward and backpropagation by hand using NumPy matrix operations, translating gradient descent calculus into working code.
- Validated understanding of weight initialization, activation functions, and learning rate effects through direct experimentation.
- Tuned hyperparameters (layer sizes, learning rate, epochs) manually to observe their effect on convergence speed and generalization.
- Used the from-scratch implementation as a foundation for later work with TensorFlow and PyTorch, grounding framework-level usage in a solid understanding of the underlying math.

ResumeOS | ***TipTap/ProseMirror, Python, FastAPI, React*** — Nov 2025 | GitHub

- Building a resume-editing tool with a rule-based Python/PyMuPDF pipeline that deterministically reconstructs document structure — headings, bullets, multi-column entries — from uploaded PDFs, and a React/TipTap editor with a Playwright + Paged.js export pipeline that preserves pixel-accurate formatting.
- Designed a shared JSON document model connecting the FastAPI backend, TipTap-based editor, and PDF export service, keeping structure-extraction, editing, and export layers fully decoupled.

Courses & Certifications

AWS Certified AI Practitioner — Sept. 2026

Coursera | ***Applied Machine Learning***

Technical Skills

Languages: Python, JavaScript, C++, SQL, Java, HTML/CSS
Frameworks & Libraries: React, Spring Boot, Flask, Tailwind, NumPy, Fastai
Tools & Platforms: Git, GitHub, Google Cloud Platform, Vercel, Render, Linux, VS Code, Neovim